

Project Design Phase

Solution Architecture

Date	1 NOVEMBER 2025
Team ID	NM2025TMID07302
Project Name	Calculating Family Expenses using Service Now
Maximum Marks	4 Marks

Solution Architecture:

The *Calculating Family Expenses Using ServiceNow* project bridges the gap between the practical challenges families face in managing household expenses and the technology platform used to address these challenges. The solution architecture defines how ServiceNow's components are configured and integrated to deliver a functional, scalable, and user-friendly system.

Purpose of the Solution Architecture:

- Identify a robust configuration within ServiceNow that can efficiently manage household financial data and automate recurring expense calculations.
- Illustrate how tables, relationships, and workflows interact to form a unified and logical data management model.
- Define clear modules and automation rules that simplify the development and maintenance process.
- Ensure the design remains extendable for future upgrades such as budget forecasting or AI-based financial insights.

Solution Structure:

The proposed architecture includes the following core components:

Core Tables:

- **Family Expenses:** Stores summary details of all family-level spending, including monthly budget, total expenses, and remaining balance.
- **Daily Expenses:** Records individual transactions (food, rent, utilities, etc.) that link directly to the corresponding family record.

Relationships:

- Establishes a one-to-many link between *Family Expenses* and *Daily Expenses* tables.
- Enables automatic data synchronization, allowing totals and summaries to update dynamically.
- Supports related list views within forms to display connected daily entries under the main family record.

Automation Components:

- **Business Rules:** Automate total expense updates, generate alerts for budget limits, and trigger notifications for overspending.
- **Number Maintenance:** Creates unique identifiers with prefixes (e.g., FEX, DEX) to differentiate tables and maintain consistency.
- **Flow Designer:** Handles real-time workflows like sending notifications or recalculating totals after new expense entries.

User Interface Elements:

- **Custom Forms:** Designed for simple and intuitive data entry with grouped fields for categories, amounts, and dates.
- **Related Lists:** Show linked daily transactions directly on the Family Expense form for easy reference.
- **Dashboards and Reports:** Provide visual summaries of monthly spending patterns, category-wise expenses, and remaining budgets.

Configuration Management:

- **Update Sets:** Maintain version control of configurations and allow seamless migration between instances.
- **Security Roles:** Assign access levels for different family members to ensure privacy and data integrity.

Phases of Development:

- **Instance Setup:** Activate and configure the ServiceNow Developer Instance.
- **Database Design:** Create Family and Daily Expense tables, define fields, and set relationships.
- **Automation Development:** Build and test Business Rules, Flow Designer processes, and Number Maintenance configurations.
- **Interface Customization:** Design forms, related lists, and dashboards for a smooth user experience.
- **Testing and Optimization:** Validate data accuracy, automation efficiency, and user accessibility.
- **Documentation and Delivery:** Prepare final reports, diagrams, and system documentation.

Specifications:

- Fully implemented using ServiceNow's low-code/no-code development environment.
- Optimized for modular design, making it adaptable to new financial categories or automation rules.
- Complies with ServiceNow's best practice framework for scalability and data integrity.
- Ready for extension with advanced analytics or mobile accessibility in future iterations.

Solution Architecture Diagram:

